Simulating and Analyzing the Real Time Data of Airport
Term Project

Subject:

Industrial Simulation

Submitted to:

Dr. Muhammad Sajid

Submitted by:

Muhammad Ali Siddique (21-IE-13)
Muhammad Ali Raza(21-IE-21)
Syed Zunair Gilani (21-IE-39)

DEPARTMENT OF INDUSTRIAL ENGINEERING
UNIVERSITY OF ENGINEERING AND TECHNOLOGY,
TAXILA

Knowledge Component	Domain	PLO mapped	Taxonomy Level	Contribution	Max Marks	Obtained Marks
APPLY the simulation concepts to model systems of practical nature	Cognitive	2	C-3	67%	5	
ANALYZE the input data by using ARENA Input Analyzer to find out the best distribution fit.	Cognitive	2	C-4		5	
USE ARENA or SIMIO software to model the problem	Cognitive	5	C-3		5	
EVALUATE the performance of the simulation model to optimize the solution	Cognitive	2	C-5		5	
ESPONDING positively towards deadline for project submissions.	Affective	11	A2	33%	5	
ESPONDING the aspects of innovation in projects	Affective	12	A2		5	

Contents

1. Introduction 4

1.1 An overview of the organization and the primary issue at hand 4

1.2 A brief description of the selected experimental design and simulation method 4

1.1.1. Entity 4

1.1.2. Process 5

Conceptual Flow Model 7

2. Methodology 7

2.3 Arena Model 8

3. Results 9

3.1 Output from Simulation 9

4. Sensitivity Analysis 9

Society 10

Environment 11

Industry 11

Health and safety 11

Sustainability 11

5. Conclusions 11

Table of Figures

Figure 1 Conceptual Flow Model 7
Figure 2 Arena Model of BDWE Bakery 8
Figure 3 Output from Simulation 9

Tables

Table 1. Entity Schedule 4
Table 2 Processing time for all Processes 5
Table 3 Resource 6
Table 4 Process Resources 6
Table 6 Results 9
Table 7 Average Utilization 10

1. Introduction

1.1 An overview of the organization and the primary issue at hand.

Given the global population's development and expansion, a discernible rise in demand has ensued, subsequently resulting in increased costs. To address the escalating costs effectively, the imperative for organizational optimization arises. Organizational optimization entails the utilization of engineering tools to explore diverse approaches aimed at optimizing resource allocation and maximizing their efficacy within the organization.

1.2 A brief description of the selected experimental design and simulation method

The selected experimental design involves studying the production process of BDWE Bakery. The bakery operates for a period of nine hours, commencing at 6 am, during which time the production of the three types of bread occurs continuously. However, despite the extended operational hours, the bakery experiences inefficiencies in terms of time consumption and productivity, with workers being relatively unoccupied during their shifts. Furthermore, the bakery faces an increased demand for its products, necessitating a need to enhance overall productivity.

The simulation method will be employed to analyze the existing production system and identify areas for improvement. By modeling the bakery's processes and simulating various scenarios, it will be possible to evaluate different strategies for optimizing productivity, reducing time consumption, and effectively meeting the increased product demand.

1.3 Overview of BDWE Bakery's Production Process

The study revolves around examining the production process of BDWE Bakery, a bakery that specializes in three types of bread: Rye Bread, Poppy seed Rolls, and Rye-wheat Bread. The bakery operates a single production line to manufacture these three distinct bread types. The production line comprises ten distinct processes, including tasks such as loading powder into the mixing machine, ingredient mixing, unloading powder from the mixing machine, dough formation, fermentation, loading bread into the baking machine, the baking process itself, and unloading bread from the baking machine, cooling, and packaging.

1.1.1. Entity

The entities arrival time starts from 6:00 pm. The amount of all powder enters to mixing process per one time is 5kg which can produce 100 breads. Rye Bread and Poppy Seed Rolls are kind of steaming bread. Rye-wheat Bread is kind of burning bread. Tunnel oven (Steaming machine) is use for steaming bread, and Rack oven (Burning machine) use for backing bread.

Table 1. Entity Schedule

Entity ID	Entity name	Start time	Arrival time(minute)
Entity 1	Rye Bread Powder	6:00am	Constant (60)

Entity 2	Poppy seed Rolls Powder	6:10am	Constant (60)
Entity 3	Rye-wheat Bread Powder	6:25am	Constant (60)

1.1.2. Process

For Mixing process, there is one mixing machines, load and unload process we use only one worker, at forming process for Rye Bread need forming by manual process which has one worker, and another 2 breads could form by Forming Machine. To produce Rye wheat Bread we have to retard the powder (fermentation) in processing time of UNIF (25,30) minutes and for another bread need to proof in processing time of UNIF (10,15). At Baking process, there are two workers for package the bread. The types of machine that we can use depend on the needed: Rack oven and Tunnel Oven. At packaging process, there are two packers to do the packaging of the bread after it already cooling UNIF (10,15) min.

Table 2 Processing time for all Processes

Process ID	**Process Name**	**Processing Time Entity 1 (Minute)**	**Processing Time Entity 2 (Minute)**	**Processing Time Entity 3 (Minute)**
P1	Load Powder to Mc	TRAI (1,2,4)		
P2	Mixing Process	UNIF (5,10)		
P3	Unload Powder from Mc	TRAI (1,1,3)		
P4	Forming Process	UNIF (30,40)	UNIF (10,15)	UNIF (10,15)
P5	Fermentation Process	UNIF (10,15)	UNIF (10,15)	UNIF (25,30)
P6	Load Bread to Backing Mc	TRAI (2,3,4)		
P7	Backing Process	CON (20)	CON (20)	CON (23)
P8	Unload Bread from Backing Mc	TRAI (1,1,3)		
P9	Cooling Process	UNIF (10,15)		
P10	Packaging Process	UNIF (10,15)		

Table 3 Resource

Resource ID	Resource name	Capacity
R1	Load Unload Operator	1
R2	Mixing machine	2
R3	Forming Staff	1
R4	Forming machine	1
R5	Proofer Machine	1
R6	Rack oven	2
R7	Tunnel oven	1
R8	Packer	2

Table 4 Process Resources

Process ID	Process name	Resource ID
P1	Load Unload Operator	R1
P2	Mixing machine	R1
P3	Load Unload Operator	R1
P4	Forming Staff	R3
P4	Forming machine	R4
P5	Proofer Machine	R5
P6	Load Unload Operator	R1
P7	Rack oven	R6
P7	Tunnel oven	R7
P8	Load Unload Operator	R1
P9	Cooling	No
P10	Packer	R8

Conceptual Flow Model

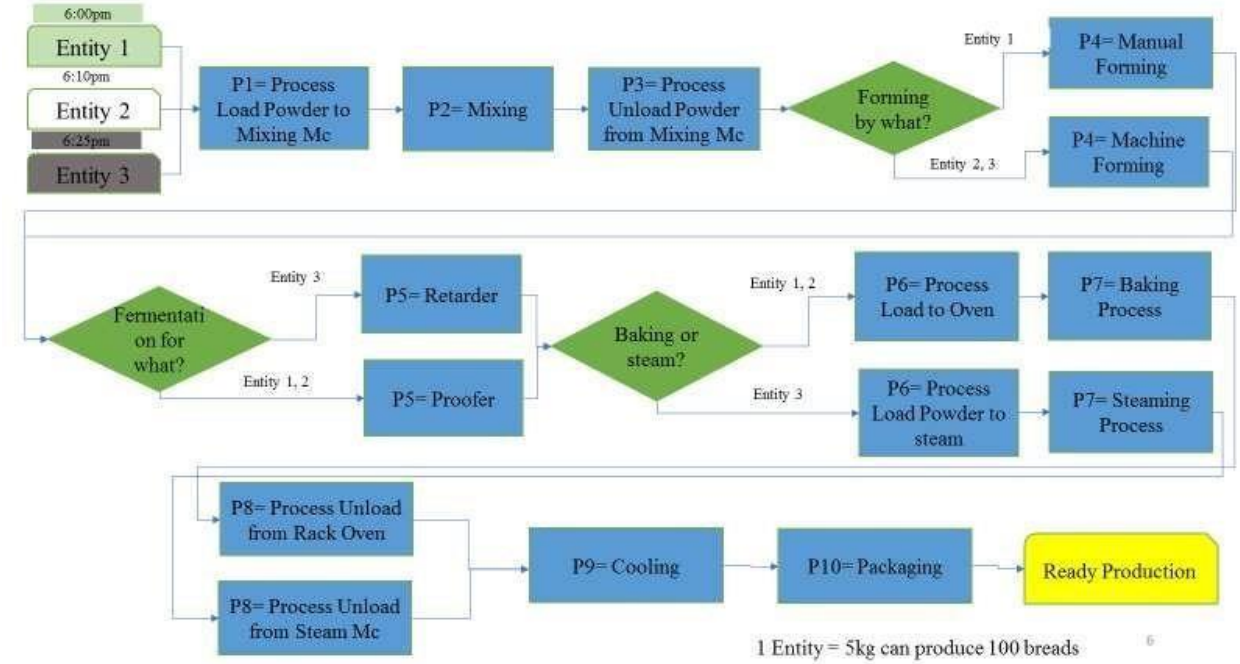


Figure 1 Conceptual Flow Model

2. Methodology

The overarching goal of this project is to optimize and boost the production output of BDWE Bakery. The specific objectives are to increase production as well as resource utilization to meet increasing demand, which includes following scenarios:

Step 1: Identification of the running parameters.

The first step is to identify the running parameters that we want to analyze:

Once we have identified the running parameters that we want to analyze, we need to decide how to measure them. For example, we may be able to use data from the equipment's control system, or we may need to collect data manually.

Step 2: Collection of data on utilization for each set of running parameters.

Once we have identified the running parameters that we want to analyze, we need to collect data on utilization for each set of running parameters. This data can be collected manually, through an automated system, or a combination of both.

In our case data was collected online.

Step 3: Analyzation of the data to identify any trends or patterns.

Once we have collected data on utilization, we need to analyze the data to identify any trends or patterns. We can use a variety of statistical methods to analyze the data, such as regression analysis, correlation analysis, and time series analysis.

The goal of the data analysis is to identify any relationships between the running parameters and utilization. For example, we may find that utilization is higher for longer running parameters, or that utilization is loour for machines that are run on multiple shifts per day.

Step 4: Development of recommendations for improving utilization based on analysis.

Once we have analyzed the data and identified any trends or patterns, we need to develop recommendations for improving utilization based on our analysis. For example, we may recommend running the equipment for longer periods of time or reducing the number of shifts per day.

When developing recommendations, it is important to consider the practical constraints of the production environment. For example, it may not be possible to run the equipment for longer periods of time if there is a high demand for product variety.

2.2. Running Parameter

- Base time unit is minute for running this simulation model.
- Run time: The run time of this model is set at 24 hours, 10 days, and 30 days.

2.3 Arena Model

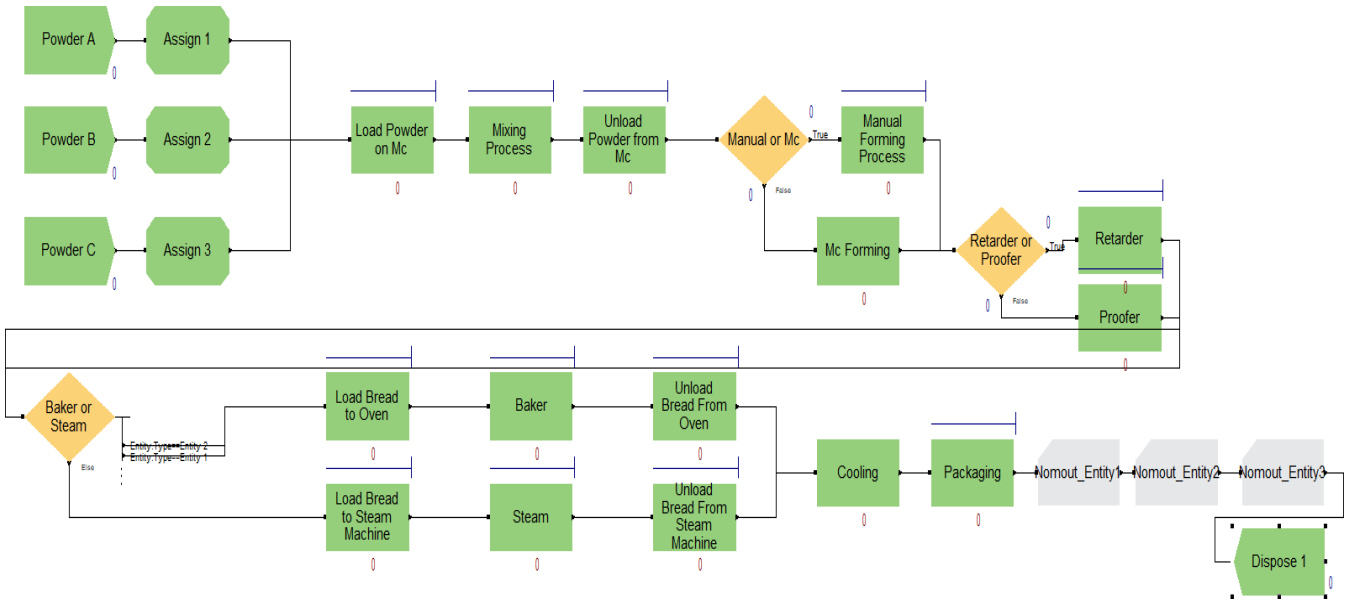
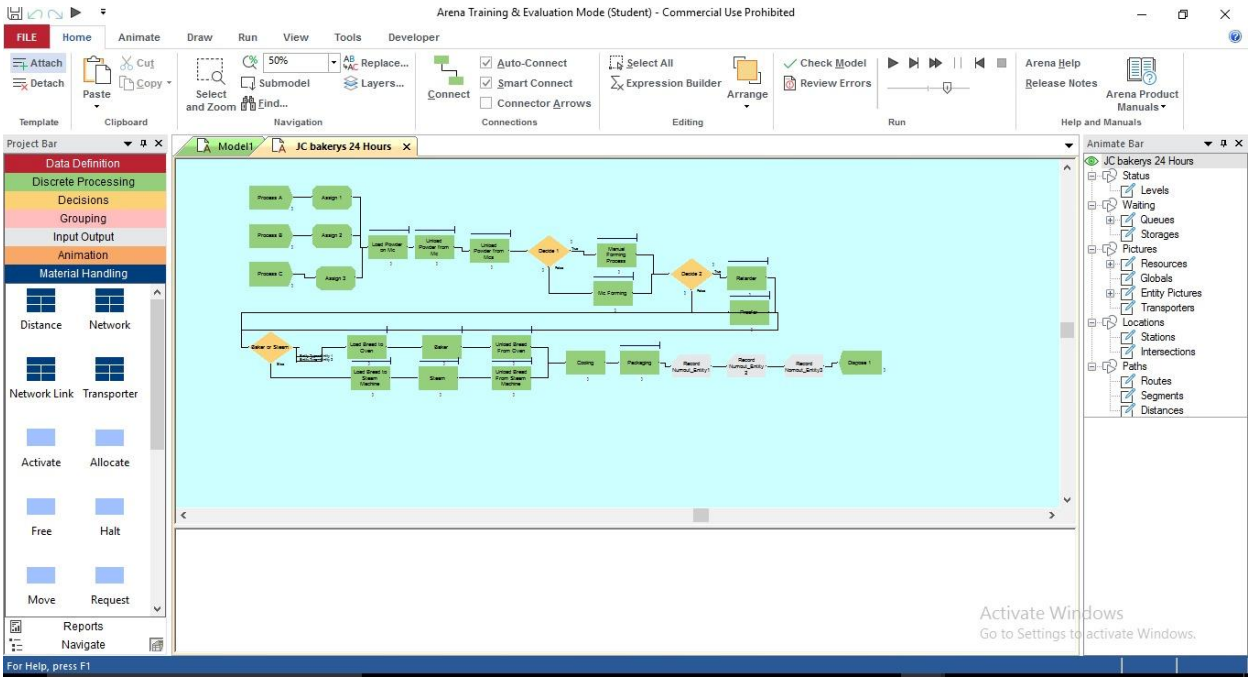
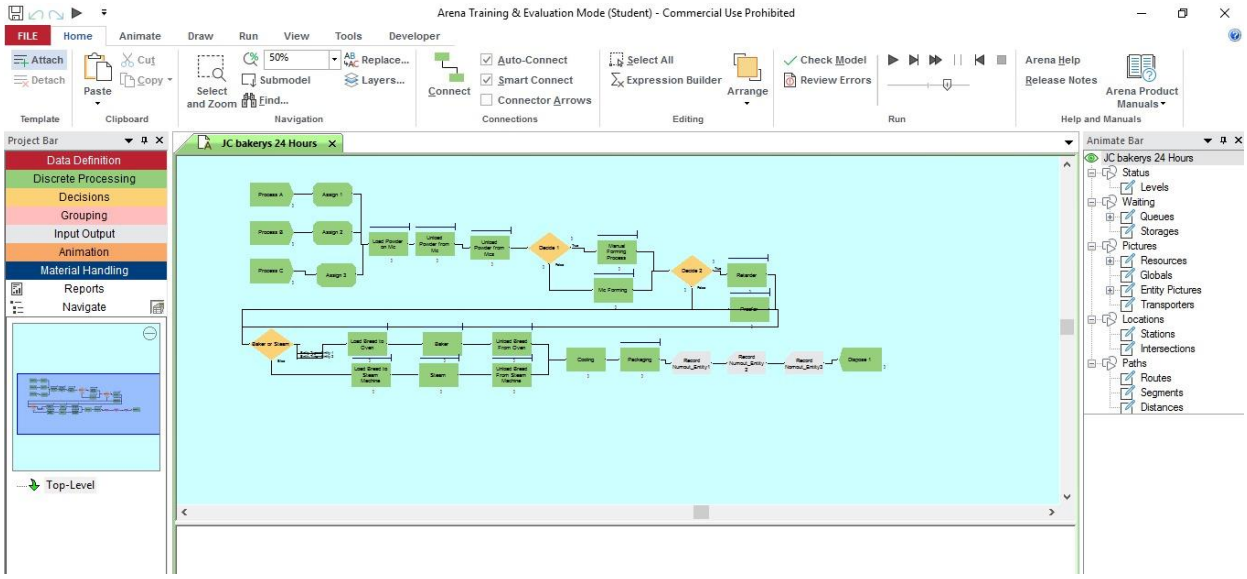


Figure 2 Arena Model of BDWE Bakery

3. Results


Arena Training & Evaluation Mode (Student) - Commercial Use Prohibited
FILE
Home
Animate
Draw
Run
View
Tools
Developer
Project Bar
Data Definition
Discrete Processing
Decisions
Grouping
Input Output
Animation
Material Handling
Distance
Network
Network Link
Transporter
Activate
Allocate
Free
Halt
Move
Request
Reports
Navigate
Model1
JC bakerys 24 Hours
Animate Bar
JC bakerys 24 Hours
Status
Levels
Waiting
Queues
Storages
Pictures
Resources
Globals
Entity Pictures
Transporters
Locations
Stations
Intersections
Paths
Routes
Segments
Distances
Activate Windows
Go to Settings to activate Windows.
For Help, press F1


Arena Training & Evaluation Mode (Student) - Commercial Use Prohibited
FILE
Home
Animate
Draw
Run
View
Tools
Developer
Project Bar
Data Definition
Discrete Processing
Decisions
Grouping
Input Output
Animation
Material Handling
Reports
Navigate
Top-Level
JC bakerys 24 Hours
Animate Bar
JC bakerys 24 Hours
Status
Levels
Waiting
Queues
Storages
Pictures
Resources
Globals
Entity Pictures
Transporters
Locations
Stations
Intersections
Paths
Routes
Segments
Distances

3.1 Output from Simulation

Discrete-Time Statistics (Tally)

Project Name	Name	Type	Source	Average Of Replication Averages	Observed-Requested	Half-Width	StDev Of Replication Averages	Minimum Replication Average	Maximum Replication Average	Overall Min Value	Overall Max Value	Avg Observations Per Replication
Unnamed Project	Bread A	NVA Time	Entity	0.047039602	Yes	0.004558845	0.000507404	0.046680813	0.047398391	0.033575965	0.062330684	22.5
		Other Time	Entity	0	Yes	0	0	(	(	(	(	22.5
		Total Time	Entity	1.517123148	Yes	0.896356864	0.099765434	1.446578333	1.587667963	1.317988185	2.354164654	22.5
		Transfer Time	Entity	0	Yes	0	0	(	(	(	(	22.5
⊟	⊟	VA Time	Entity	1.392226784	Yes	0.051095697	0.005687003	1.388205466	1.396248103	1.278503009	1.508535208	22.5
		Wait Time	Entity	0.077856762	Yes	0.849820013	0.094585835	0.010974476	0.144739047	(	1.01753714	22.5
	Bread B	NVA Time	Entity	0.049621345	Yes	0.024281485	0.002702554	0.04771035	0.051532339	0.037981996	0.066166859	14
		Other Time	Entity	0	Yes	0	0	(	(	(	(	14
		Total Time	Entity	1.728667481	Yes	1.323371418	0.14729259	1.624515891	1.83281907	1.336498007	2.568032882	14
		Transfer Time	Entity	0	Yes	0	0	(	(	(	(	14
	⊟	VA Time	Entity	1.370335202	Yes	0.100211543	0.011153647	1.362448382	1.378222021	1.277593585	1.458507415	14
		Wait Time	Entity	0.308710934	Yes	1.447864447	0.161148791	0.194761531	0.422660337	(	1.20895645	14
	Load Bread to Steam Machine.Queue	Waiting Time	Queue	0.001294326	Yes	0.007626776	0.000848868	0.000694086	0.001894566	(	0.021569662	38.5
	Load Powder on Mc.Queue	Waiting Time	Queue	0.002816765	Yes	0.00115192	0.00012821	0.002726106	0.002907423	(	0.035716141	40.5
	Mc Forming.Queue	Waiting Time	Queue	0.019984107	Yes	0.099698898	0.011096589	0.012137633	0.02783058	(	0.28562445	39.5
	Packaging.Queue	Waiting Time	Queue	0	Yes	0	0	(	(	(	(	37
	Proofer.Queue	Waiting Time	Queue	0.004844696	Yes	0.036064286	0.004013992	0.002006375	0.007683017	(	0.068550651	38.5
	Record Nomout_Entity3	Expression	User Specified	0	Yes	0	0	(	(	(	(	36.5
	Record Numout_Entity 2	Expression	User Specified	4.50719697	Yes	19.92082629	2.217208309	2.939393939	6.075	(	16	36.5
	Record Numout_Entity1	Expression	User Specified	13.24280303	Yes	2.315031999	0.257665426	13.06060606	13.425	(	23	36.5
	Steam.Queue	Waiting Time	Queue	0.133187623	Yes	0.823160469	0.091618601	0.068403489	0.197971757	(	1.114106962	36
	Unload Bread From Steam Machine.Queue	Waiting Time	Queue	0.003269002	Yes	0.002072301	0.000230649	0.003105909	0.003432096	(	0.025713378	37
	Unload Powder from Mc.Queue	Waiting Time	Queue	0.010129549	Yes	0.042486911	0.004728837	0.006785757	0.013473342	(	0.114972456	39.5
	Unload Powder from Mcs.Queue	Waiting Time	Queue	0.00043488	Yes	0.003619946	0.000402904	0.000149984	0.000719776	(	0.0214574	39.5

Figure 3 Output from Simulation

4. Sensitivity Analysis

Table 5 Results

Parameter	Value	Running Parameter
Load Unload Operator.ScheduledUtilization	.14460 or 14.460%	24 hours
Forming Machine.ScheduledUtilization	.33679 or 33.679%	24 hours
Packer.ScheduledUtilization	.31706 or 31.706%	24 hours
Load Unload Operator.ScheduledUtilization	.24311 or 24.311%	10 days
Forming Machine.ScheduledUtilization	.58361 or 58.361%	10 days
Packer.ScheduledUtilization .51723	.51723 or 51.723%	10 days

Load Unload Operator.ScheduledUtilization	.24154 or 24.154%	30 days
Forming Machine.ScheduledUtilization	.58210 or 58.210%	30 days
Packer.ScheduledUtilization .51723	.51496 or 51.496%	30 days

Table 6 Average Utilization

Parameter	Average utilization
Load Unload Operator.ScheduledUtilization	20.98%
Forming Machine.ScheduledUtilization	50.08%
Packer.ScheduledUtilization	44.98%

As shown in the data, the utilization is higher for the 10-day and 30-day running parameters than for the 24-hour running parameter. This suggests that the equipment is more efficiently utilized when it is running for longer periods of time.

Reasons:

- Setup and changeover time: It takes time to set up and change over equipment for different products or jobs. When the equipment is running for longer periods of time, this setup and changeover time is spread out over a larger number of units, which reduces the overall impact on utilization.
- Efficiency gains: Equipment often becomes more efficient as it is used for longer periods of time. This is because the operators become more familiar with the equipment and learn how to operate it more efficiently. Additionally, the equipment may warm up and reach its optimal operating temperature, which can also improve efficiency.
- Reduced downtime: When the equipment is running for longer periods of time, there is less opportunity for downtime. This is because there are feour times when the equipment needs to be shut down for maintenance or repairs.

Running equipment for longer periods of time can lead to significant efficiency gains and reduced downtime, which can have a positive impact on utilization.

Society

Increasing the number of forming machines could lead to more jobs in the manufacturing sector. Decreasing the number of forming staff members could lead to job losses. Increasing the number of proofer machines and tunnel ovens could lead to increased production of goods, which could benefit consumers. Increasing the number of load/unload operators could improve their utilization, which could lead to increased productivity and loour costs. Changing the configuration settings of a simulation model could also have an impact on the quality of life for people living near manufacturing facilities. For example, if the model is used to simulate the production of air pollutants, then changing the settings could lead to changes in air quality.

Environment

Increasing the number of forming machines, proofer machines, and tunnel ovens could lead to increased energy consumption and greenhouse gas emissions. Changing the configuration settings of a simulation model could also have an impact on water usage and waste production.

Industry

Changing the configuration settings of a simulation model could lead to changes in the way that goods are produced. For example, if the model is used to simulate a manufacturing process, then changing the settings could lead to changes in the types of machines and equipment that are used. Changing the configuration settings of a simulation model could also have an impact on the profitability of businesses. For example, if the model is used to simulate the production of a product, then changing the settings could lead to changes in the cost of production and the selling price of the product.

Health and safety

Changing the configuration settings of a simulation model could have an impact on the health and safety of workers. For example, if the model is used to simulate a manufacturing process, then changing the settings could lead to changes in the types of hazards that workers are exposed to.

Sustainability

Changing the configuration settings of a simulation model could have an impact on the sustainability of a business or organization. For example, if the model is used to simulate the production of a product, then changing the settings could lead to changes in the environmental impact of the product.

5. Conclusion

Through a comprehensive analysis of the bakery model, the impact of various factors has been determined. To ensure efficiency and reliability of the system, an Arena Simulation model was developed and multiple scenarios oure evaluated using process analyzer. Based on the insights gained from the Arena model and process analyzer, it is formally concluded that augmenting the quantity of proofer machines contributes to the enhancement of production levels. Furthermore, reducing the number of forming staff members enhances their utilization, and increasing the count of load/unload operators also improves their utilization within the system. These findings provide valuable guidance for optimizing the bakery operations and maximizing productivity.